



## Your maxon drive

consists of:

### **Planetary gearhead GPX 22 Ø22 mm, 1-stage**

Configurable

### **DCX 22 S Ø22 mm, Precious Metal Brushes, ball bearings**

Configurable

### **Encoder ENX 10 EASY, 1 - 1024 CPT**

Configurable

## Your Contact

maxon motor worldwide

[http://www.maxongroup.com:80/maxon/view/content/contact\\_page](http://www.maxongroup.com:80/maxon/view/content/contact_page)

E-Mail: [info@maxongroup.com](mailto:info@maxongroup.com)

Internet: <http://www.maxongroup.com:80>

retail price:

|                       |                   |
|-----------------------|-------------------|
| <b>0-4 Pieces</b>     | <b>€374.82</b>    |
| <b>5-19 Pieces</b>    | <b>€321.57</b>    |
| <b>20-49 Pieces</b>   | <b>€273.25</b>    |
| <b>from 50 Pieces</b> | <b>On Request</b> |

## Terms and conditions

## Planetary gearhead GPX 22 Ø22 mm, 1-stage

### Configurable



#### General information

|                |                  |
|----------------|------------------|
| Gearhead type  | GPX              |
| Outer diameter | 22 mm            |
| Version        | Standard version |

#### Gearhead Data

|                                         |                         |
|-----------------------------------------|-------------------------|
| Reduction                               | 6.6 : 1                 |
| Absolute reduction                      | 66/10                   |
| Number of stages                        | 1                       |
| Max. continuous torque                  | 0.5 Nm                  |
| Max. intermittent torque                | 0.6 Nm                  |
| Direction of rotation, drive to output  | Equal                   |
| Max. efficiency                         | 90 %                    |
| Average backlash no load                | 0.85 °                  |
| Mass inertia                            | 0.5908 gcm <sup>2</sup> |
| Gearhead length (L1)                    | 19.9 mm                 |
| Max. transmittable power (continuous)   | 24 W                    |
| Max. transmittable power (intermittent) | 30 W                    |

#### Technical Data

|                               |                               |
|-------------------------------|-------------------------------|
| Bearing at output             | Waelzlager                    |
| Radial play                   | max. 0.2 mm, 5 mm from flange |
| Axial play                    | 0 - 0.2 mm                    |
| Max. radial load              | 65 N, 10 mm from flange       |
| Max. axial load (dynamic)     | 40 N                          |
| Max. force for press fits     | 100 N                         |
| Max. continuous input speed   | 8000 rpm                      |
| Max. intermittent input speed | 10000 rpm                     |
| Recommended temperature range | -40...+100 °C                 |
| Number of autoclave cycles    | 0                             |

#### Product

|        |      |
|--------|------|
| Weight | 45 g |
|--------|------|

## DCX 22 S Ø22 mm, Precious Metal Brushes, ball bearings

### Configurable



#### Values at nominal voltage

|                                           |          |
|-------------------------------------------|----------|
| Nominal voltage                           | 12 V     |
| No load speed                             | 6200 rpm |
| No load current                           | 19.6 mA  |
| Nominal speed                             | 4670 rpm |
| Nominal torque (max. continuous torque)   | 14.7 mNm |
| Nominal current (max. continuous current) | 0.817 A  |
| Stall torque                              | 59.7 mNm |
| Stall current                             | 3.25 A   |
| Max. efficiency                           | 85 %     |

#### Characteristics

|                          |                       |
|--------------------------|-----------------------|
| Terminal resistance      | 3.69 $\Omega$         |
| Terminal inductance      | 0.231 mH              |
| Torque constant          | 18.4 mNm/A            |
| Speed constant           | 520 rpm/V             |
| Speed / torque gradient  | 104 rpm/mNm           |
| Mechanical time constant | 6.07 ms               |
| Rotor inertia            | 5.55 gcm <sup>2</sup> |

#### Thermal data

|                                    |              |
|------------------------------------|--------------|
| Thermal resistance housing-ambient | 16 K/W       |
| Thermal resistance winding-housing | 7 K/W        |
| Thermal time constant winding      | 20 s         |
| Thermal time constant motor        | 528 s        |
| Ambient temperature                | -40...+85 °C |
| Max. winding temperature           | +100 °C      |

#### Mechanical data

|                                    |                                                                |
|------------------------------------|----------------------------------------------------------------|
| Bearing type                       | Waelzlager                                                     |
| Max. speed                         | 7160 rpm                                                       |
| Axial play                         | 0 mm, at radial load < 2.5 N<br>0.1 mm, at radial load > 2.5 N |
| Radial play                        | 0.02 mm                                                        |
| Max. axial load (dynamic)          | 2.5 N                                                          |
| Max. force for press fits (static) | 30 N                                                           |
| (static, shaft supported)          | 440 N                                                          |
| Max. radial load                   | 16 N, 5 mm from flange                                         |

#### Other specifications

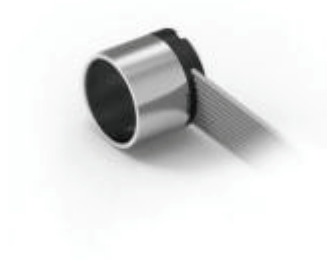
|                               |   |
|-------------------------------|---|
| Number of pole pairs          | 1 |
| Number of commutator segments | 9 |
| Number of autoclave cycles    | 0 |

#### Product

|        |      |
|--------|------|
| Weight | 66 g |
|--------|------|

## Encoder ENX 10 EASY, 1 - 1024 CPT

### Configurable



#### General information

|                                |           |
|--------------------------------|-----------|
| Counts per turn                | 1024      |
| Number of channels             | 3         |
| Line Driver                    | RS422     |
| Max. length of encoder housing | 8.5 mm    |
| Max. electrical speed          | 90000 rpm |
| Max. mechanical speed          | 30000 rpm |

#### Technical Data

|                                               |                          |
|-----------------------------------------------|--------------------------|
| Supply voltage Vcc                            | 5.0V &plusmn; 10.0%      |
| Output signal                                 | EIA RS 422               |
| Output signal                                 | Incremental              |
| Driver used logic                             | Differential, EIA RS 422 |
| Output current per channel                    | -20...20 mA              |
| Signal rise time                              | 20 ns                    |
| Measurement condition for signal rise time    | CL=25pF, RL=1kOhm        |
| Signal fall time                              | 20 ns                    |
| Measurement condition for signal fall time    | CL=25pF, RL=1kOhm        |
| Min. state duration                           | 125 ns                   |
| Direction of rotation                         | A before B CW            |
| Index position                                | A low & B low            |
| Index synchronized to AB                      | Yes                      |
| Typical current draw at standstill            | 22 mA                    |
| Max. moment of inertia of code wheel          | 0.05 gcm <sup>2</sup>    |
| Operating temperature                         | -40...+100 °C            |
| Orientation of encoder output to motor flange | 360 °                    |

#### Product

|        |     |
|--------|-----|
| Weight | 5 g |
|--------|-----|